

Multi-Modal Highlight Detection in Broadcast Audio

A Deep Learning Approach for Event Recognition in Sports and
eSports

Nuno M. C. da Costa

DTx -- Digital Transformation CoLAB (RTO) - Applied Research - University of Minho, Portugal

Rise of Live Streaming

-  **Explosive Growth:** Sports and eSports content consumption is surging globally.
-  **Modern Demands:** Audiences require near real-time analysis and instant summaries.
-  **Workflow Challenge:** This pressure is pushing the limits of traditional broadcast workflows.

The Challenge & The Need

-  **Annotation Costly & Slow:** Manual annotation cannot keep up with the volume of content.
-  **Visual Limitations:** Visual-only systems fail due to occlusions and replay transitions.
-  **Critical Need:** We require scalable, automated, and lightweight solutions to improve User Experience.

Why Audio?



Emotional Cues

Audio captures the raw emotion of the event. Crowd reactions (cheering, gasping) and commentator tone (shoutcasting) provide distinct signals of importance.



Efficiency

Audio processing is computationally far lighter than video analysis (edge deployment). It is also robust to visual artifacts like camera cuts, occlusions, and replay transitions.



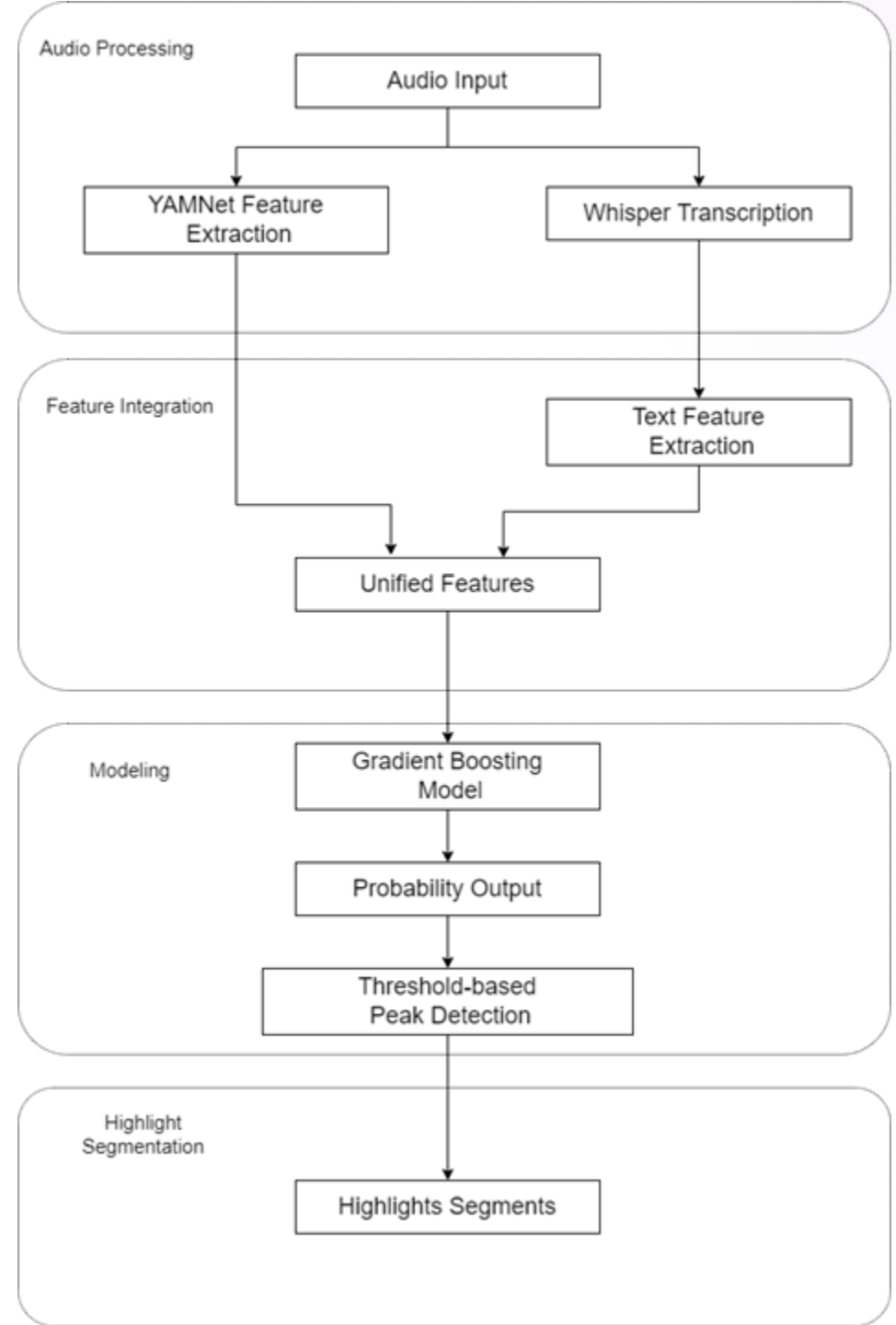
Objective

The goal is to create a lightweight, low-latency system that complements existing video analysis pipelines or functions autonomously for rapid content generation.

The Multi-Modal Architecture

Our proposed architecture fuses high-level semantic features from two distinct pathways:

- **Audio Path:** Utilizes a pretrained YAMNet model to extract 16 semantic classes from the raw audio stream.
- **Text Path:** Employs OpenAI's Whisper for ASR and DeBERTa (Zero-Shot) to classify commentary sentiment.
- **Fusion:** A Gradient Boosting classifier combines these signals to output a final probability time series.



Feature Extraction: Audio (YAMNet)

YAMNet Model

 **Key Technology:** We utilize **YAMNet**, a deep net pretrained on the massive AudioSet corpus.

-  **Robust Understanding:** This model provides robust, high-level semantic understanding.
-  **Efficiency:** Eliminates the need for manual feature engineering.

16 Semantic Classes

From 521 classes, we selected 16 relevant to event detection, including:

- > **Crowd:** Cheering, Applause, Chanting
- > **Vocal:** Speech, Shouting, Whoop
- > **Specifics:** Whistling, Air horn, Siren

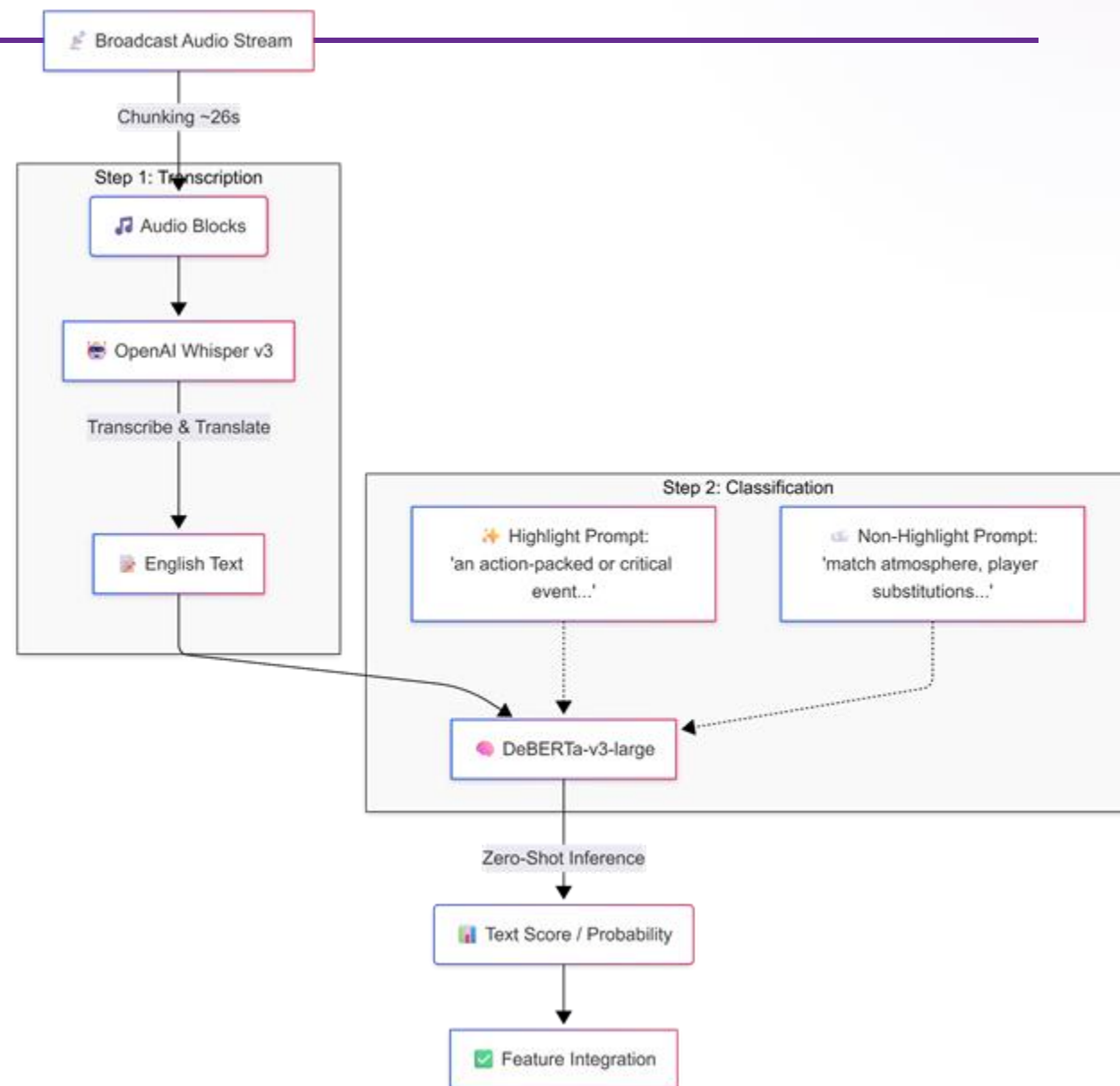
Signals are downsampled to ~4.8s time bins.

Feature Extraction: Text (NLP)

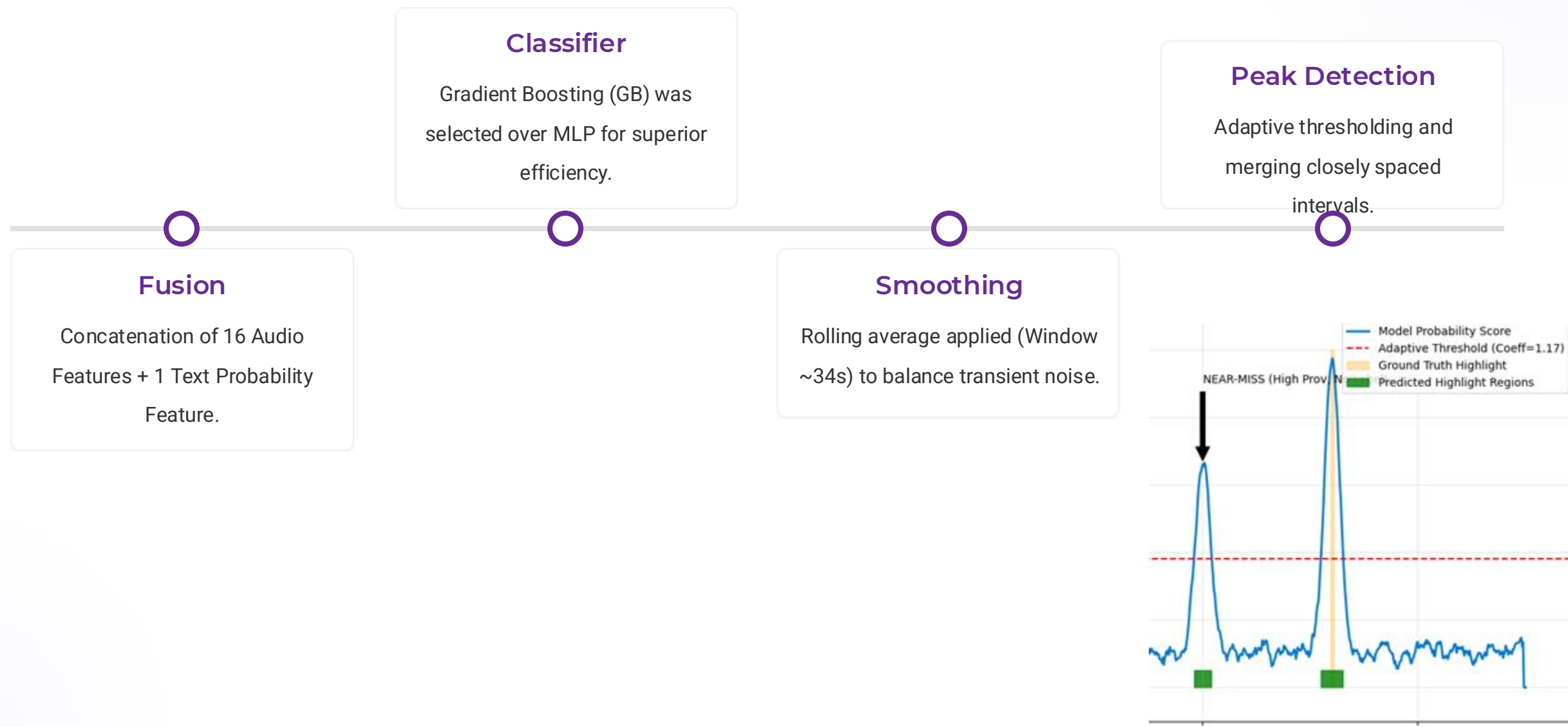
The Semantic Layer

Commentary provides distinct semantic cues often missing from raw audio energy. Our pipeline operates in two steps:

- **1. Transcription:** Utilizing OpenAI's Whisper (v3) to transcribe and translate commentary into English in ~26s blocks.
- **2. Classification:** DeBERTa-v3-large (Zero-Shot) assigns a probability score based on prompts like "an action-packed event" vs. "match atmosphere".



Classification & Post-Processing



Dataset & Experimental Setup

SoccerNet-v2

550 Games: 500 Training / 50 Test (European Leagues).

Label Engineering

We condensed the raw 17 classes into a binary target:

Highlight vs. Non-Highlight.

Temporal Expansion: To capture the "buildup + reaction" cycle, we expand the anchor label to include the preceding bin and two subsequent bins (-1 to +2), creating a ~17.8s window.

Custom Operational Metrics (Viewer Experience):

Coverage: How concise is the reel?

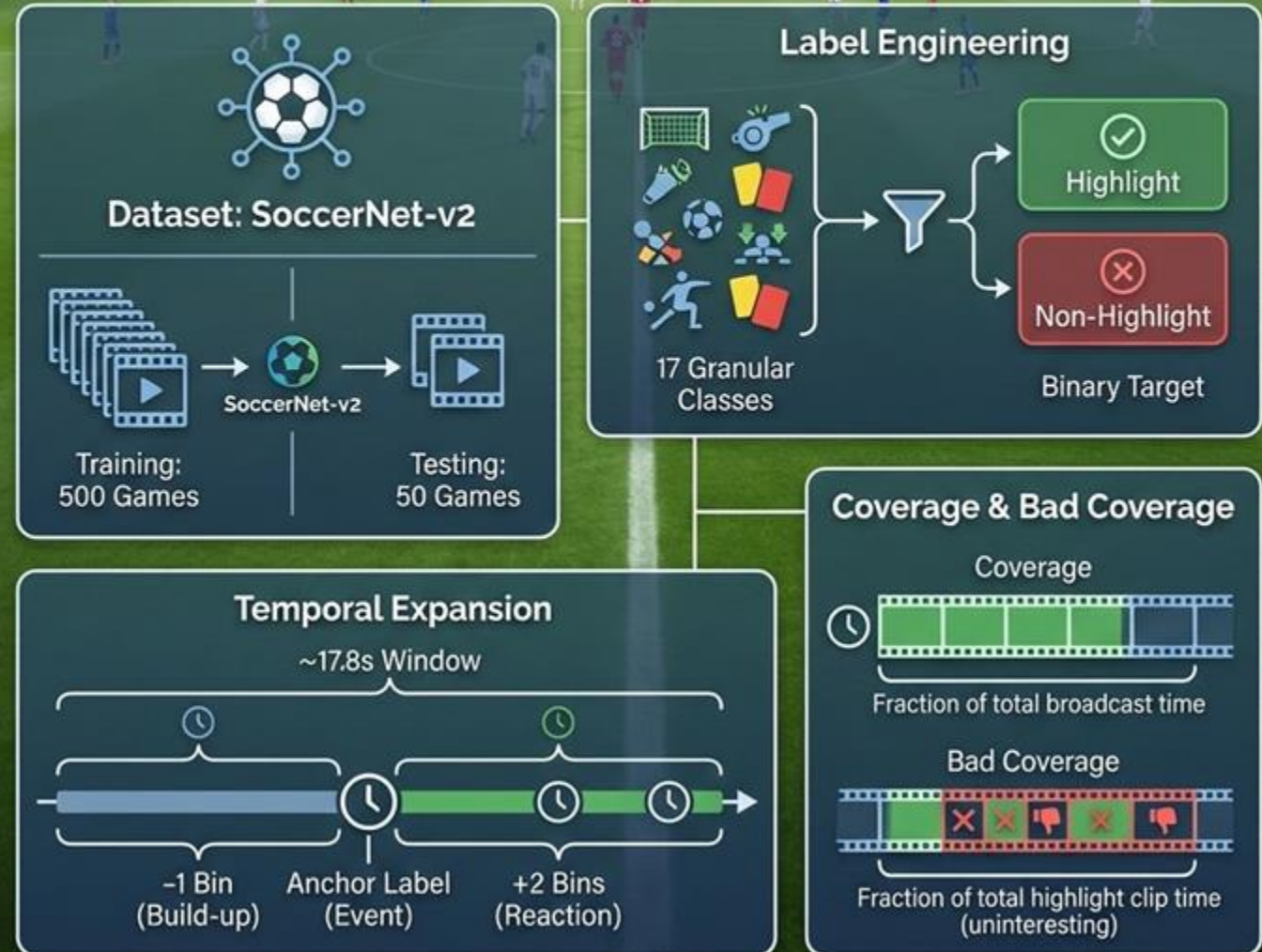
(Highlight Time / Total Broadcast Time) → Low

Bad Coverage: How much fluff is in the reel?

(Uninteresting Time / Total Highlight Time) → ~0

Dataset & Experimental Setup

How we trained and validated.



Results: Labeling and Features

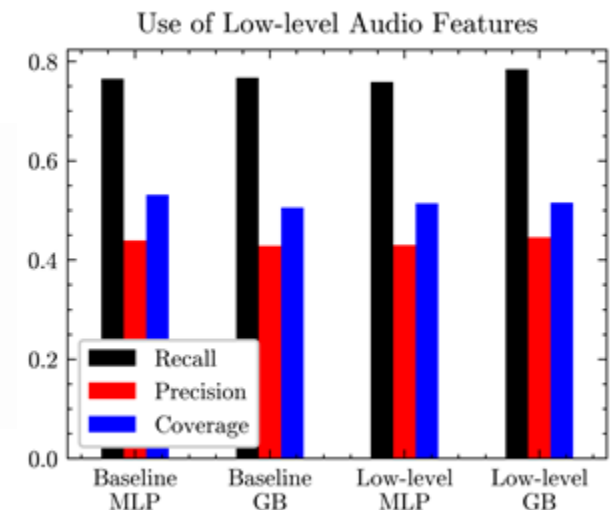
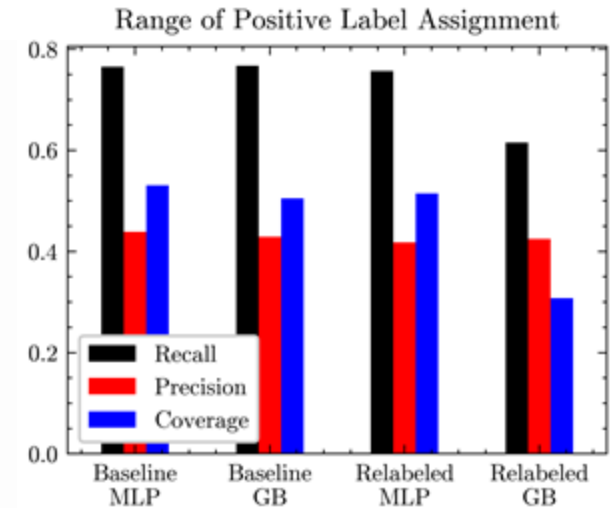


Labeling Engineering: compared **wider** (2 preceding, 4 subsequent bins) vs. **concentrated** (1 preceding, 2 subsequent bins, ~17.8s) strategies.

Concentrating positive labels **reduces overall Coverage, ensuring a concise summary**. Tighter labeling eliminates ambiguous audio, enhancing class separation and **minimizing Bad Coverage (uninteresting fluff)**, leading to more precise highlight detection via Gradient Boosting and adaptive thresholding.



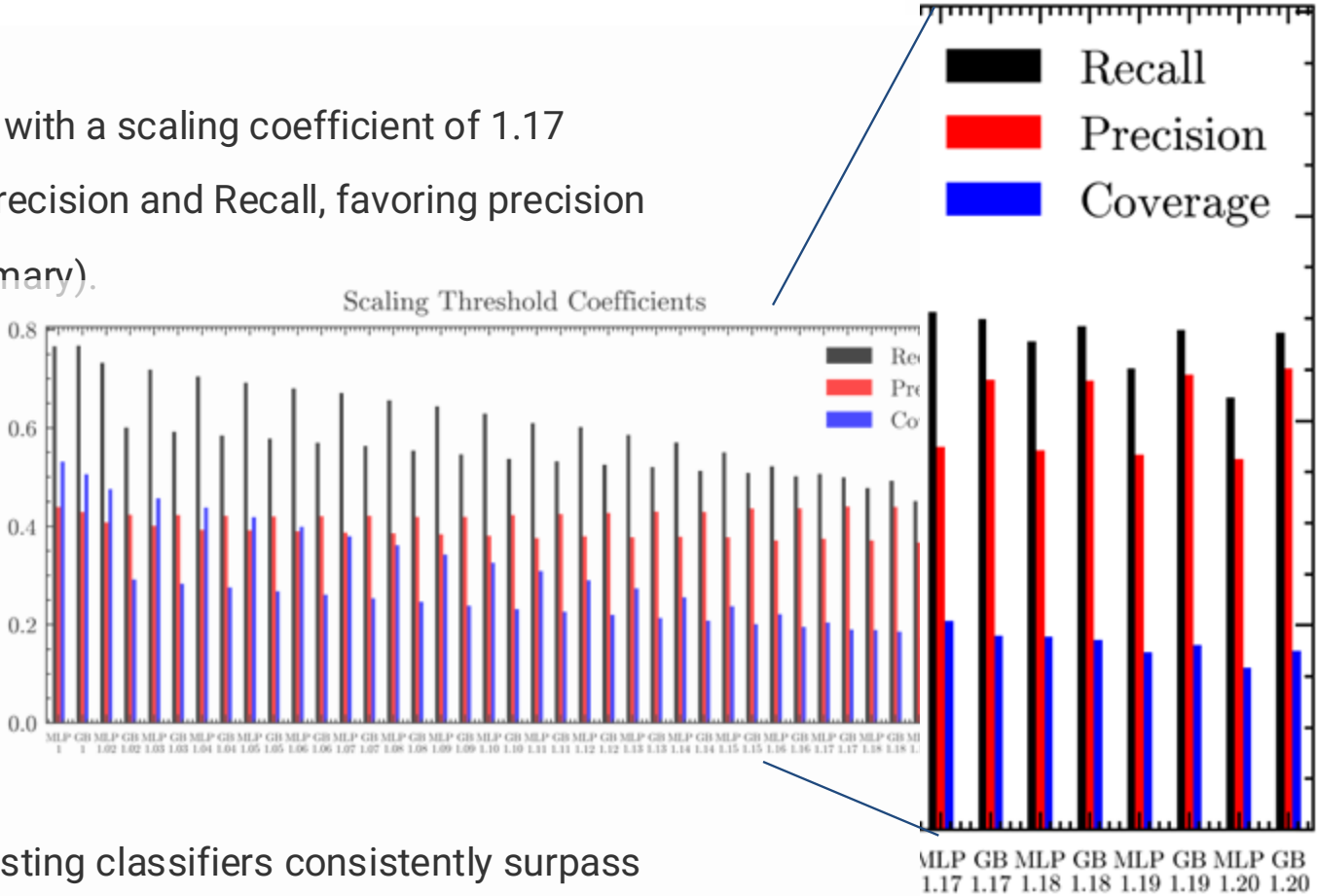
High-Level vs. Low-Level Features: Impact of incorporating low-level audio features (Pitch, Loudness), in addition to selected YAMNET scores and a text feature. **Low-level features increased inference cost without yielding consistent performance improvements**, validating our semantic approach.



Results: Thresholding and Model Selection



Threshold Tuning: An adaptive threshold with a scaling coefficient of 1.17 provided the optimal trade-off between Precision and Recall, favoring precision and reducing the coverage (concise summary).



Gradient Boosting vs. MLP: Gradient Boosting classifiers consistently surpass Multi-Layer Perceptrons on validation subsets for thresholds within this range of low coverage, particularly regarding precision.

Results: Quantitative Benchmark

Quantitative held out tests on 3 matches: Impact of Multi-Modal Fusion on Detection Performance

Model Configuration	F1-Score	Precision	Recall	Coverage
Audio Only (YAMNet)	39.3%	33.0%	48.8%	20.3%
Audio + NLP (Proposed)	46.8%	44.0%	49.9%	19.0%

Key Takeaway: Adding NLP improves F1 by **+7.5%** and Precision by **+11%** while keeping low coverage.

Results: Qualitative Manual error Analysis

Reasons for manual error analysis

1. **Label Noise:** SoccerNet anchors often miss subjective, human-perceived highlights.
2. **Meaningful False Positives:** Manual inspection reveals our model consistently flags exciting, unannotated plays (e.g., dangerous near-misses).
3. **Visual Confirmation:** Probability signals perfectly align with genuine peaks in excitement.
4. **Adjusted error analysis:** by manually disregarding on confusion matrix False Negative classifications for events that upon manual review, were deemed uninteresting, even if they were labeled as highlights in the SoccerNet dataset.

Manual adjusted error analysis

Model Configuration	F1-Score	Precision	Recall	Coverage	Bad Coverage
Audio + NLP (adjusted)	54.5%	47.6%	63.8%	19.4%	28.1%

Key Takeaway: Our model outperforms Soccernets dataset's human annotators.

*Factoring in unannotated highlights, our **"True" Semantic Precision is ~72%** (47.6% Baseline + 24.3% Meaningful False Positives).*

24.3% calculated as: $100\% - 47.6\% [\text{false positives}] - 28.1\% [\text{Bad Coverage}]$

Limitations & Future Work

Beyond Soccer

The proposed architecture is highly generalizable. Future work involves transfer learning to specific eSports titles like League of Legends or CS:GO, which share similar "shoutcasting" and crowd dynamics.

Personalization

We aim to implement tunable thresholds, allowing users to select their preferred density of highlights (e.g., "Show me only goals" vs. "Show me all shots").



Conclusion

 **Multi-modal Fusion Works:** Combining semantic audio scores with commentary transcription significantly outperforms audio-only baselines.

 **Efficiency:** Pretrained features and shallow classifiers enable low-latency inference suitable for live streaming applications.

 **Simplicity:** High-level semantic features remove the need for complex, brittle, and manually handcrafted acoustic engineering.

 **Generality:** The architecture is promising for any domain with commentary and event-driven audio, including the rapidly growing eSports sector.

Q&A

Thank you for your attention.

Nuno M. C. da Costa

nuno.costa@dtx-colab.pt

DTx -- Digital Transformation CoLAB

This work was supported under the base funding project of the DTx CoLAB - Collaborative Laboratory, under the **Missão Interface of the Recovery and Resilience Plan (PRR)**, integrated in the notice 01/C05-i02/2022. We thank our Portuguese industry partners for their collaboration.

